Ethan Mader

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EDUCATION

Computer Science Ph.D., Purdue | Aug 2024 - Present | West Lafayette, IN
GPA: 3.94/4.0; focus on theoretical computer science, advised by Kent Quanrud

Computer Science B.S., Mathematics B.S., UCSB | Sep 2020 - June 2024 | Goleta, CA
GPA: 3.93/4.0; **SAT**: Math 800; **SAT Math II**: 800; Theta Tau Professional Engineering Fraternity

SKILLS: Python (Numpy, Pandas, CVX, Matplotlib), C++, SQL, Convex Optimization, Linear Programming, Linear Algebra, Graph Algorithms, Randomized Algorithms

AWARDS & HONORS

- **Presidential Excellence PhD Award**: $40,000 over 4 years given to the top 1.5% of incoming PhD students at Purdue in 2024
- **Herbold Scholarship**: Selected to receive $10,000 out of 200+ CS PhD students at Purdue
- **UCSB Dean's Honors**: Awarded for 11 of 12 academic quarters
- UCSB *magna cum laude* in College of L.S., *cum laude* in CoE

RESEARCH

Research Assistant, Purdue | Sep 2024 - Present | West Lafayette, IN
- Improved upon randomized techniques to make runtime improvements for the classic edit distance problem under the relaxation to a 3-approximation
- Summarized state-of-the-art graph decomposition techniques to optimize the efficiency of flow-based algorithms on large networks

Undergrad Researcher, UCSB Programming Languages Lab | Jan - June 2023 | Goleta, CA
- Developed novel techniques for algorithmically generating mathematical proofs via code, enhancing formal verification techniques
- Research findings led to the translation of a theorem-proving package from Agda to Coq for use in circuit verification

TEACHING

Math and CS Tutor, Wyzant | June 2023 - Sep 2024 | CA
- Remote instruction on competition math and C++ fundamentals, data structures, and algorithms
- 200+ hours and perfect 5-star rating across 30 reviews, with one-time and long-term clients

PROJECTS

Matrix Tensor Factorization
- Used alternating least squares with graduated regularization and symmetry constraints in Python to obtain low rank matrix tensor factorizations
- Recovered Strassen's algorithm, and found new exact factorizations for multiplying 3 by 3 matrices

Kalah Bot
- Created a 3D interactable game tree for the board game Kalah with 12-ply state evaluation in Python via minimax with alpha beta pruning
- Studied the game tree to develop a nearly unbeatable Kalah bot for player 1 which uses simple human-understandable heuristics and only looks 2 moves deep

INTERESTS: Jane Street monthly puzzles, bullet chess (top 2% on chess.com), rock climbing, tennis, hiking